

Siddharth Hiraou

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 2+ years of experience designing, deploying, and optimizing scalable machine learning and Generative AI solutions across cloud environments. Expertise in Python, TensorFlow, PyTorch, LLMs, RAG, LangChain, MLOps, and AWS, delivering production-ready AI systems that improve model accuracy, deployment efficiency, and business outcomes. Strong background in deep learning, NLP, feature engineering, and end-to-end ML lifecycle management, with a proven record of accelerating deployment cycles, enhancing prediction performance, and building enterprise-grade AI applications using modern cloud-native architectures.

SKILLS

Programming Languages: Python, Java, C++, R, SQL, Bash Scripting

AI/ML Frameworks & Libraries: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers

Machine Learning Techniques: Supervised Learning, Unsupervised Learning, Deep Learning, NLP, Computer Vision, Reinforcement Learning, Time Series Forecasting, Model Fine-Tuning

Generative AI / LLM: LLM Integration, Prompt Engineering, RAG (Retrieval-Augmented Generation), LangChain, OpenAI API, Fine-Tuning LLMs, Vector Databases (Pinecone, FAISS, ChromaDB)

MLOps & Deployment: MLflow, Docker, Kubernetes, CI/CD Pipelines, Model Versioning, Model Monitoring, Airflow, Kubeflow

Cloud Platforms: AWS (SageMaker, EC2, S3, Lambda), Azure ML, Google Cloud Platform (Vertex AI, BigQuery ML)

Data Engineering & Processing: ETL Pipelines, Apache Spark, Hadoop, Data Wrangling, Feature Engineering, Data Pipeline

Databases: SQL, NoSQL, MongoDB, PostgreSQL, Snowflake

Visualization & Analytics Tools: Power BI, Tableau, Matplotlib, Seaborn

Version Control & Collaboration: Git, GitHub, GitLab, Jira, Agile/Scrum

Soft Skills: Cross-functional Collaboration, Stakeholder Communication, Problem Solving, Research & Innovation

EXPERIENCE

Avnet, NY | AI/ML Engineer

Jan 2026 – Present

- Architect scalable machine learning pipelines using Python, PyTorch, MLflow, and Docker, reducing model deployment time by 34% while improving release reliability across production environments.
- Develop Retrieval-Augmented Generation (RAG) applications with LangChain, vector databases, and LLMs, increasing response accuracy by 28% for enterprise knowledge search.
- Optimize deep learning models through hyperparameter tuning, feature engineering, and distributed training, improving prediction performance by 19% while lowering inference latency by 24%.
- Implement automated MLOps workflows leveraging Git, CI/CD, Kubernetes, and AWS services, decreasing model release cycles by 31% and minimizing deployment errors.
- Integrate NLP solutions using Hugging Face Transformers and TensorFlow for document intelligence, boosting entity extraction precision by 23% and reducing manual review efforts.
- Collaborate with cross-functional stakeholders to deliver AI-driven analytics using SQL, Python, and cloud data platforms, accelerating business decision-making by 27% through production-ready ML solutions.

Siemens, India | ML Engineer

Sep 2022 – Jul 2024

- Designed, supervised, and unsupervised machine learning models using Scikit-learn and Python, increasing predictive accuracy by 21% across manufacturing analytics initiatives.
- Engineered robust data preprocessing and feature engineering pipelines with Pandas and NumPy, improving training data quality by 26% and reducing preparation time by 30%.
- Evaluated multiple classification and regression algorithms through cross-validation and performance benchmarking, improving model selection efficiency by 18%.
- Automated model training and validation workflows using Python scripting and SQL, decreasing repetitive manual effort by 32% while ensuring consistent experiment tracking.
- Enhanced neural network performance with TensorFlow optimization techniques, reducing training duration by 22% and improving inference consistency across production datasets.
- Validated model outputs through statistical analysis, error monitoring, and performance reporting, increasing prediction reliability by 17% and supporting data-driven engineering decisions.
- Delivered machine learning solutions alongside software and business teams, enabling analytics adoption that improved operational process efficiency by 20% across targeted business functions.

EDUCATION

Master of Science in Data Science

Jan 2026

University at Buffalo, The State University of New York, Buffalo, NY

Bachelor of Engineering - BE, Information Technology

Jun 2024

Savitribai Phule Pune University, India

CERTIFICATION

Amazon Web Services (AWS)